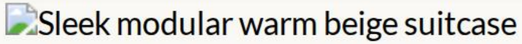
Flow Unit 01

Journey Unbound: Reclaiming the Joy of Travel



"Travel is Joyful, but..."

Identifying the hidden bottlenecks that disrupt the flow of experience.

Defining the "Pauses"

Physical Friction

Heavy, rigid luggage restricts movement and causes fatigue during transitions between destinations.

Auditory Pollution

The constant rattle of hard wheels on urban cobblestones creates mental stress and public noise.

Companion Isolation

Traveling with pets requires separate, bulky carriers that further complicate the logistics of "flow."

The Solution: All-in-One System

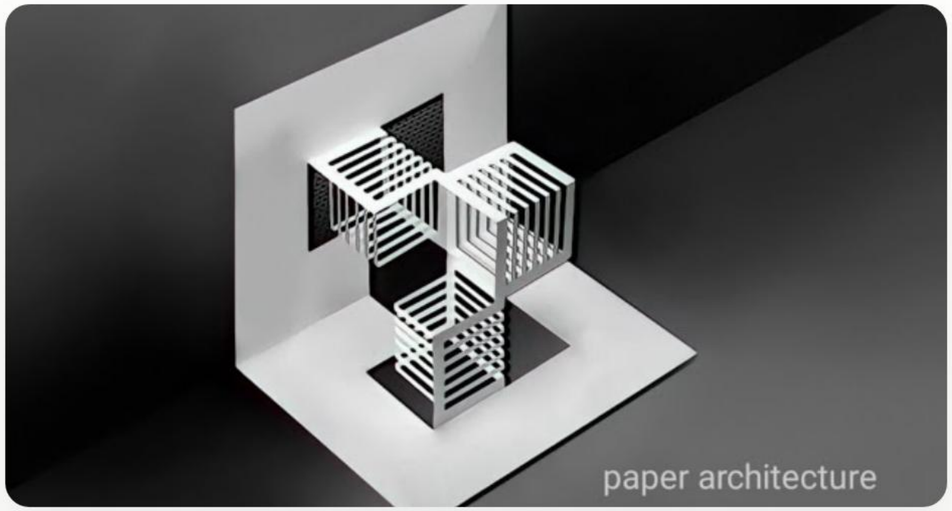
A Seamless Transition

Flow Unit 01 integrates every travel need into a single, modular entity. It is not just a bag or a carrier; it is a companion that adapts to your journey's rhythm.

- **Hybrid Carry:** Bag mode to Carrier mode.
- **Pet Integration:** External modular docking.
- **Furniture Mode:** Ground-docking chair.

Retractable wheels mechanism

The Kinetic Structure



Origami Expansion

Inspired by Miura-ori, the "Accordion" structure allows for a massive 40% volume expansion while maintaining structural rigidity for cargo safety.

Kirigami Flexibility

Strategic laser-cut patterns enable the unit to bend and curve around physical obstacles, maintaining movement even in cluttered urban environments.

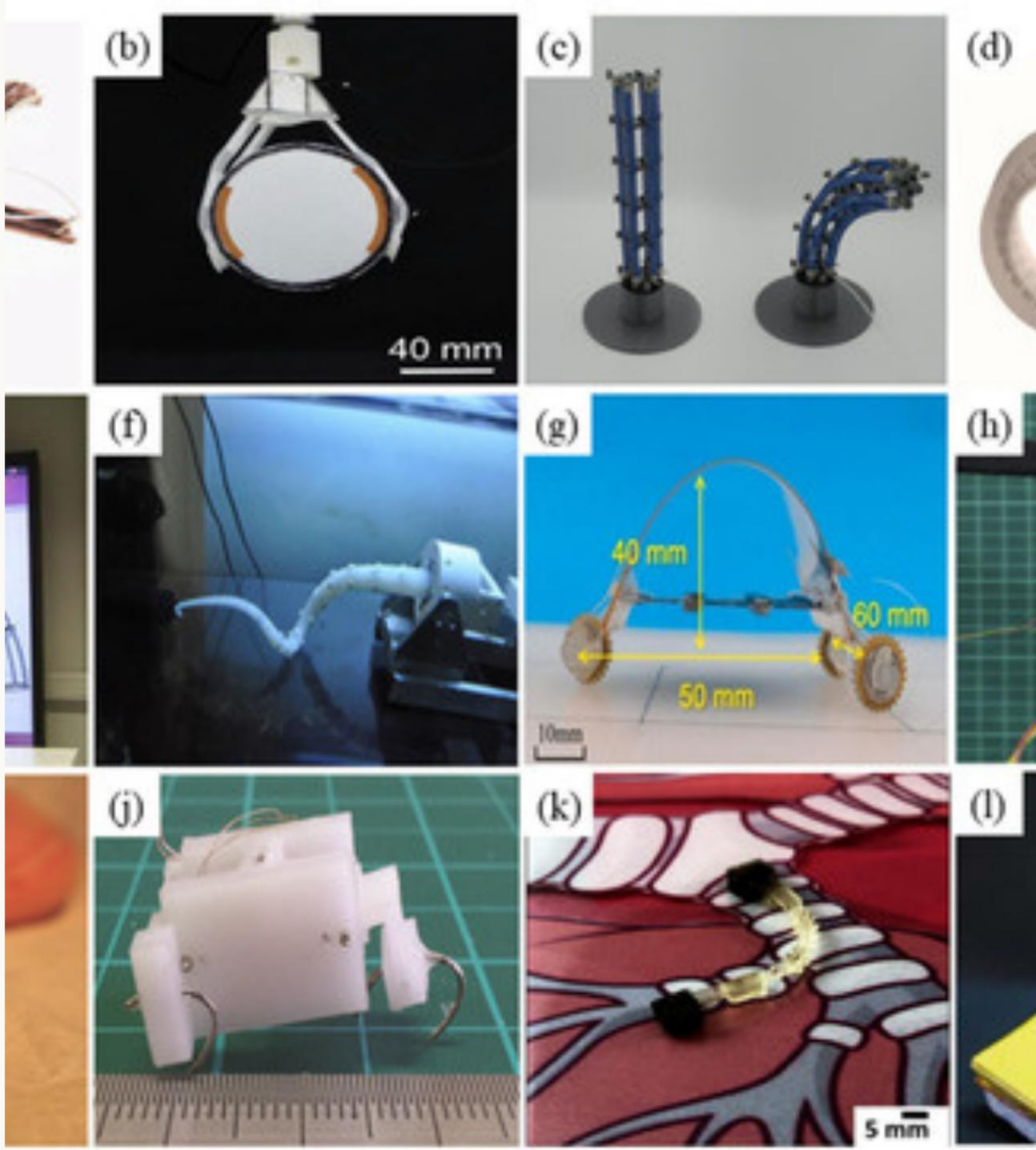
Soft-Robotic Wheels

Silent & Retractable

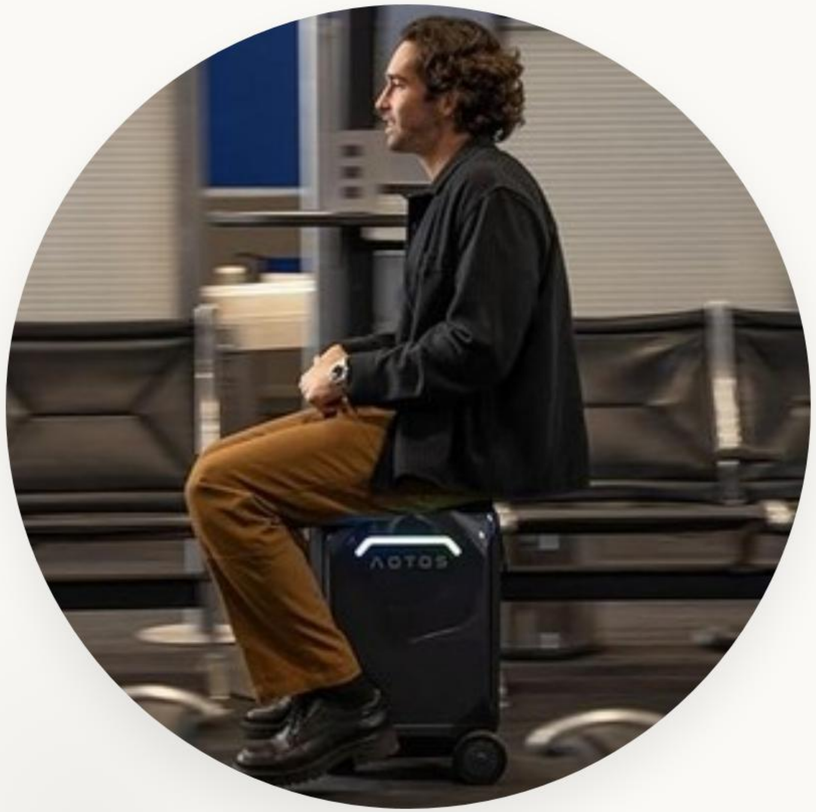
Utilizing soft-pneumatic actuators, the wheels adjust their **stiffness** in real-time. This eliminates noise on rough roads and absorbs impact.

The Retraction Logic:

When unnecessary, the wheels hide completely within the body, transforming the unit into a sleek bag for handheld transport.



Furniture: The "Joyful Stop"

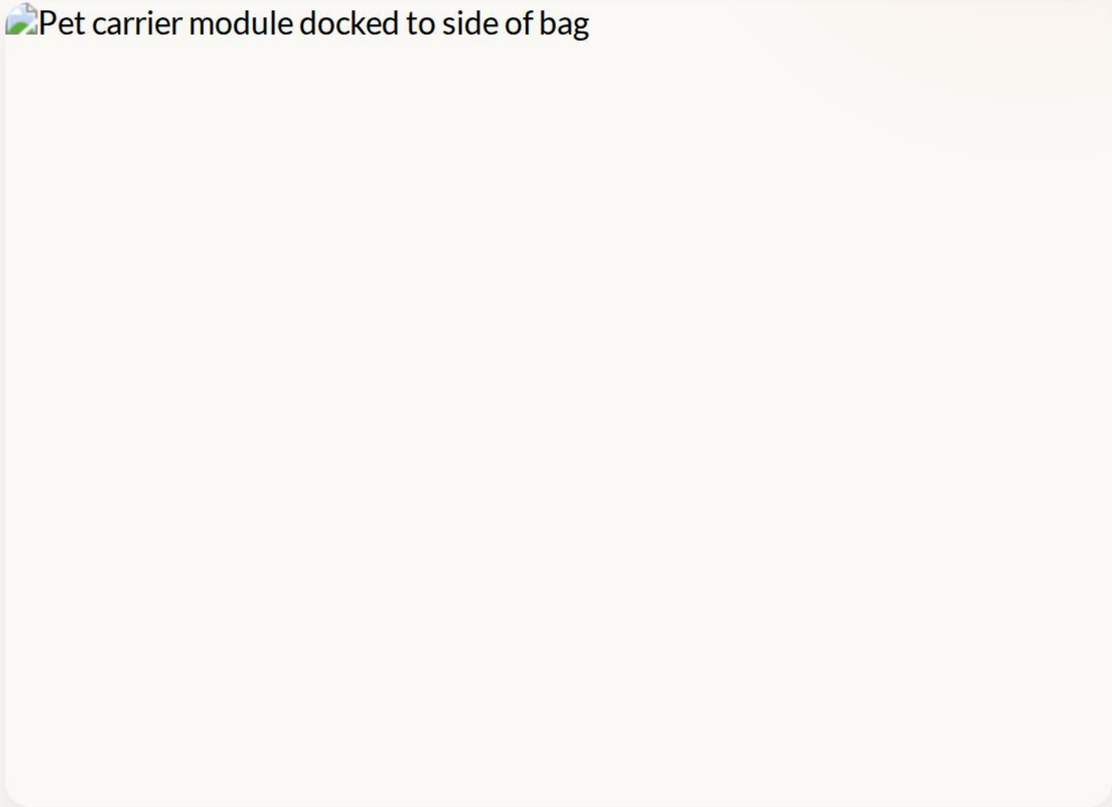


Docking & Rest

Why wait uncomfortably? At the push of a button, Flow Unit 01 lowers itself and docks to the ground. The retractable wheels disappear, providing a zero-wobble, stable chair mode for immediate rest anywhere.

$$\text{Stability}_{\text{Docked}} > 300\ \%\ \text{Stability}_{\text{Wheeled}}$$

Modular Pet Companion



Never Excluded

Traditional pets are locked in cages away from the flow. Our modular system docks the pet module to the **exterior**, allowing for visual contact and shared experience while keeping cargo separate and safe inside.

"Travel is more joyful when shared with those we love."

The Transformation Sequence

Preparation

Bag Mode: Wheels retracted. Lightweight handheld mode for quick errands.

Transit

Carrier Mode: Wheels deployed. Soft-robotic stiffness for silent urban navigation.

Waiting

Chair Mode: Unit docks to ground. Full stability for long-duration rest.

Performance Comparison

Compared to industry-leading luxury hard-shell luggage.

CMF & Sustainability

Layer	Material	Color Strategy
Outer Skin	Recycled High-Density Fabric	Warm Beige (#F5F5DC)
Kirigami Joints	Shape-Memory Alloy	Rose Gold Metallic
Soft Actuators	Medical-Grade Elastomer	Peach Orange (#FF8C69)

Flow Unit 01

Join the movement toward a frictionless future.

www.journeyunbound.design

Image Sources

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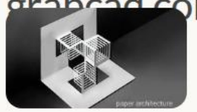

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